INCH-POUND

MIL-T-6368A(USAF)
20 July 1989
SUPERSEDING
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14 November 1950

MILITARY SPECIFICATION

TRANSMITTER, FUEL FLOW, TOTALIZING, TYPE H-1A

INACTIVE FOR NEW DESIGN AFTER 4 JUNE 1971.

This specification is approved for use by the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type of totalizing fuel flow transmitter, designated Type H-1A, to be used for indicating, remotely, the total fuel consumed by an aircraft turbo-jet, and is for use on the high pressure side of the engine fuel pump.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEOR, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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SPECIFICATIONS

Federal

L-P-378	Plastic Sheet And Strip, Thin Gauge, Polyolefin
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Method Of
DOD-D-1000	Drawing, Engineering And Associated List
MIL-C-5015	Connector, Electrical, Circular, Threaded, An Type, General Specification For
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-S-7742	Screw Threads, Standard, Optimum Selected For Series, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-810C	Environmental Test Methods
MIL-STD-970	Standards And Specifications, Order Of Preference For The Selection Of
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements
MS33678	Connector, Receptacle, Electric, Integral Mounting"

Air Force-Navy Aeronautical

AND10050	Bosses-Standard Dimensions For Gasket Seal Straight Thread

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

(Copies for MIL-STD-810C is available from Director, Naval Publications and Printing Service Office, 700 Robbins Ave, Bldg 4 Section D, Philadelphia, PA 19111.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2)."

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

The parenthetical phrase "(except for related associated detail specifications, specification sheets or MS standards)" shall be omitted from the above paragraph for specifications which do not have related associated detail specification, specification sheets, or MS standards.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Material:

3.2.1 Specifications and standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-970, except as provided in the following paragraph:

3.2.1.1 Standard parts. AN or MIL Standard parts shall be used wherever they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts such parts such as screws, bolts, nuts, cotter pins, et cetera may be used, provided they have suitable properties and are replaceable by the AN or MIL Standard parts without alteration, and provided the corresponding AN or MIL parts numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

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3.2.2 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.3 Nonferrous materials. Nonferrous materials shall be used for all parts of the transmitter, except where ferrous materials are essential.

3.2.4 Recycled and reclaimed materials. Recycle and reclaimed materials shall be encourage to the maximum extent possible without jeopardizing the intended end use of the item.

3.3 Design and construction. The transmitter shall conform to Figure 1. Unless otherwise specified, the transmitter shall be designed to transmit from a remote station, the fuel consumed or remaining in gallons of aviation gasoline in accordance with Specifications MIL-F-5572 and MIL-E-5624. The type of fuel for initial calibration shall be as specified by the Procuring Agency. The transmitter shall be of the positive displacement type. The transmitter shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations and other conditions incident to shipping, installation, and storage.

3.3.1 Case. The case of the transmitter shall be designed to be rain tight and dust tight. The transmitter case shall be constructed of aluminum, aluminum alloy, bronze, or steel, wherever it is suitable for the purpose.

3.3.2 Magnetic switch. The transmitter shall incorporate a hermetically sealed magnetically operated switch. The switch shall operate once for each gallon of fuel consumed, thus closing the circuit through the indicator.

3.3.3 Pressure. The transmitter shall withstand a maximum working pressure of 1200 psi without distortion of any part sufficient to produce errors in indication. The transmitter shall withstand a pressure of 1500 psi without bursting or taking a permanent set.

3.3.4 Flow direction. The direction of flow through the transmitter shall be permanently and legibly marked by stamped or case lettering, "In" and "Out".

3.3.5 Caps. The fuel line connections shall be furnished with suitable caps or plugs to prevent foreign material entering the transmitter.

3.3.6 Adjustments. The transmitter unit shall be provided with means to adjust or correct the indications, if this becomes necessary during the life of the transmitter. This means of adjustments shall be reasonably simple to manipulate and of such a nature that adjustment can be accomplished with tools ordinarily possessed by a depot instrument repairman. This adjustment may consist of changing gears.

3.3.7 Fuel line connections. Each fuel line connection shall be in accordance with Drawing AND10050 for 1 inch outside diameter tubing.

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3.4 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification DOD-D-1000.

3.5 Wiring. The transmitter shall be wired in accordance with the diagram shown on Figure 2.

3.6 Weight. The weight of the transmitter shall not exceed 8.5 pounds.

3.7 Electrical receptacle. The transmitter shall be provided with a pressurized electrical receptacle in accordance with MS33678-12S-3P and Specification MIL-C-5015.

3.8 Mounting studs. The length of the transmitter mounting studs from the lock nut to the end shall be a minimum of 0.56 inch.

3.9 Lock wire. All transmitter body screws and nuts shall be safetied with brass lock wire. The lock wire shall be twisted tight between each pair of screws or nuts.

3.10 Radio noise filter. A suitable filter shall be placed across the magnetic switch to reduce the radio noise.

3.10.1 Electromagnetic interference. The transmitter shall meet the electromagnetic interference requirements of MIL-STD-461 for class A1 category b equipment, except the frequency range shall be from 0.15 to 150 MHz.

3.11 Screw threads. Unless otherwise specified, the threads of all machine screws 0.06 inch or larger in diameter shall conform to Specification MIL-S-7742.

3.12 Finishes and protective coatings.

3.12.1 Aluminum alloys parts. Aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Electrical receptacles and small screws need not be anodized, but shall be suitably protected from the environmental conditions specified herein.

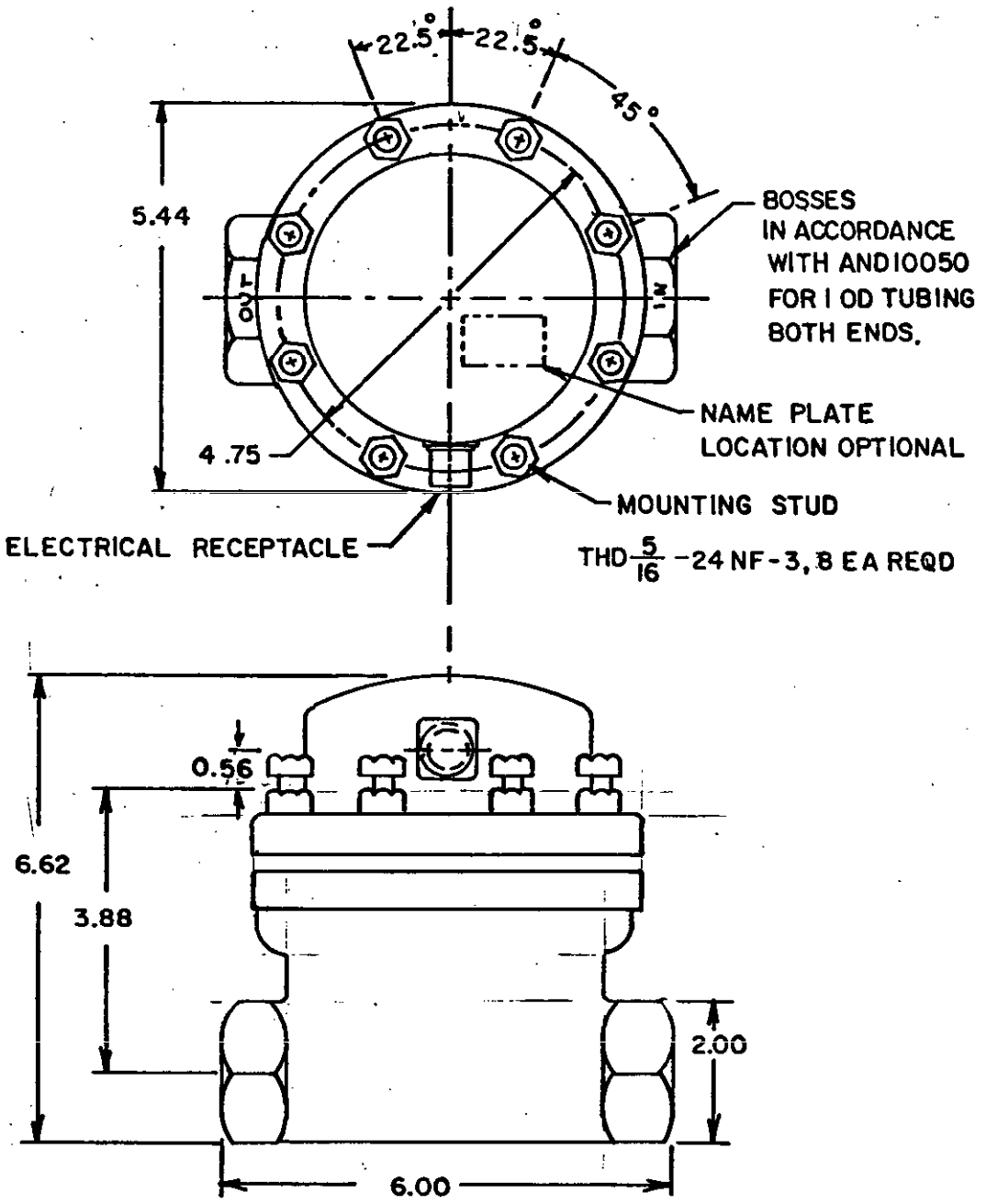
3.12.2 Iron and steel parts. Iron and steel parts shall be suitably protected to meet the Humidity tests specified herein.

3.12.3 Brass parts. Brass parts shall be finished as specified for iron and steel parts.

3.12.4 Mildew proofing. The electrical components shall be sprayed with a fungicide which will protect these components against the environmental conditions specified in Section 4.

3.13 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130."

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES : ANGLES ± 0° 15'
DECIMAL ±.06

FIGURE 1. Transmitter outline.

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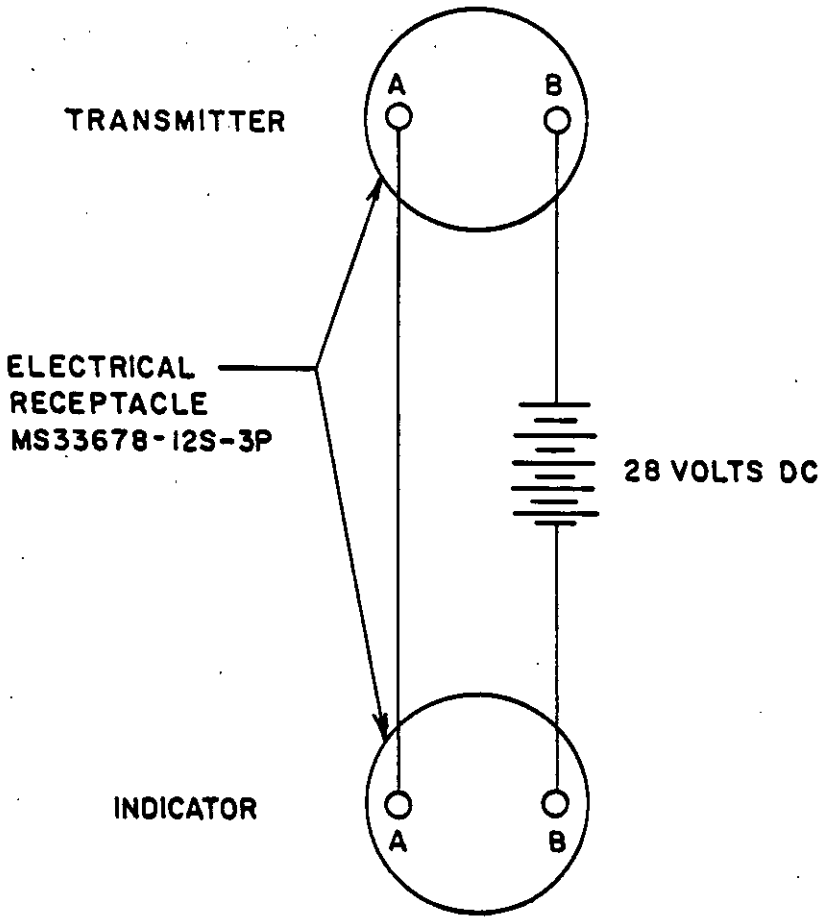


FIGURE 2. Wiring diagram.

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3.14 Wiring diagram. A wiring diagram showing external wiring between indicator, transmitter, and power source of a type acceptable to the Procuring Agency shall be painted or baked on the transmitter case. Decalcomanias may be used if protected with a suitable laquer.

3.15 Warning instructions. With each transmitter one printed or reproduced copy of the following warning instructions shall be included in a prominent manner: "USE 1-0.31-12N-3 STRAIGHT THREAD FUEL LINE FITTINGS ONLY TO CONNECT TRANSMITTER. DO NOT USE PIPE THREAD FITTINGS."

3.16 Envelope. An envelope containing the warning instructions shall be securely attached to each transmitter. The following information shall be printed on the face of the envelope:

"IMPORTANT
THIS ENVELOPE CONTAINS WARNING
INSTRUCTIONS, MOUNTING NUTS."

3.17 Workmanship:

3.17.1 General. The transmitter, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner in accordance with MIL-STD-454 requirement. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges, accuracy of dimensions, radii of fillets, marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring and riveting, alignment of parts and tightness of assembly screws, bolts, et cetera.

3.17.2 Cleaning. The transmitter shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspections requirements specified herein are classified as follows:

a. First article (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.3.1.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test shall be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from these values, proper allowance shall be made for the difference from the specified condition. Except where otherwise specified, the transmitter shall be tapped or vibrated before a test reading is taken.

4.3.2 Vibration stand. Whenever a vibration stand is specified, it shall vibrate at any desired frequency between 8 and 50 Hz and shall subject the transmitter to vibration such that a point on the transmitter case will move in a straight line with a total excursion of not less than 0.030 inch nor more than 0.040 inch, and with sinusoidal velocity. This movement shall be along any axis (x, y, or z) of a cartesian coordinate system. One axis shall be parallel to the vertical centerline of the transmitter in its normal position.

4.3.3 Position. Unless otherwise specified, the transmitter shall be tested with the common plane of the mounting studs horizontal, and with the electrical receptacle at the top.

4.3.4 Test voltage. The transmitter shall be tested with an applied voltage of 28V dc.

4.3.5 Test fuel. All transmitter tests, except the vibration test and the Substitute fuel test shall be conducted using the type of fuel specified or approved by the Procuring Agency.

4.3.6 The transmitters shall be tested with an electrical solenoid counter operated by the switch in the transmitter. Sufficient fuel shall be used in testing in order that the number of gallons for each rate of flow shall not be less than the value specified in Table I.

TABLE I. Fuel flowmeter transmitter.

True Flow Rate		Indicated	Tolerance-percent of Gallons Fuel Consumed Plus or Minus	
Gals/Hr	Lbs/Hr (Approx.)	Fuel Consumed Gallons	Room Temp	High Temp. Low Temp. and Substitute Fuel
100	700	1	1.0	2.0
400	2700	10	1.0	2.0
1200	8200	10	1.0	2.0
1500	10500	10	1.0	2.0

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4.3.7 Test pressure. Unless otherwise specified, all transmitter tests shall be conducted with a line fuel pressure between 10 psi and 50 psi.

4.4 First article inspection. Unless otherwise specified, contractor's records of all inspection work and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in the specification, and acceptable to the Government.

4.4.1 First article inspection sample tested by the contractor. The contractor shall subject a minimum of three transmitters to the first article test specified in 4.4.3.

4.4.2 First article test sample for the procuring activity. The contractor shall submit sample(s) to the contracting activity who will use it (them).

a. For a review of the mechancial construction of the product.

b. To perform any tests, included in the specification, after reviewing the contractor's test report.

4.4.3 First article inspection. The first article inspection shall consist of the following:

Test	Paragraph
a. Individual tests	4.4.4
b. Position error	4.4.5.5
c. Vibration	4.4.5.6
d. Special tests	4.4.6
e. Visual inspection	4.4.5.4

4.4.4 Individual tests. Each transmitter shall be subjected to the following tests. In addition, each transmitter shall be subjected to any other test specified herein which the Inspector considers necessary to determine compliance with the requirements of this specification.

4.4.4.1 Scale error. The transmitter shall be tested for scale errors at the points shown in Table I. The scale errors shall not exceed the tolerances specified in Table I.

4.4.4.2 Pressure drop. The transmitter shall be tested for pressure drop at the rates of flow specified in Table II. The pressure drop of the transmitter shall not exceed the values specified in column covering "Pressure Drop" of Table II.

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TABLE II

Flow Rate Gals/Hr	Flow Rate Lbs/Hr (Approx.)	Pressure Drop Pounds per Square Inch	Blocked Mechanism Pressure Drop Pounds per Square Inch
400	2700	1.0	18.0
1200	8200	7.0	22.0

4.4.4.3 By-pass valve. The metering mechanism shall be blocked by any suitable means and the pressure drop across the transmitter shall be measured with the fuel flowing at the rates specified in Table II. The pressure drop values shall not exceed the tolerances specified in column covering "Blocked Mechanism" of Table II. The by-pass valve shall not vibrate or chatter, but shall open and close positively.

4.4.4.4 Leak. The transmitter shall be tested for leaks by applying a hydralic pressure of 1200 psi to the inlet connection with the outlet connection plugged or capped. This test shall first be performed at room temperature and no leaks shall be evident over a period of 5 minutes. The temperature of the transmitter shall then be reduced to $-54 \pm 2^{\circ}C$ by submersion in a cold bath already at this temperature, for a period of not less than 15 minutes. While still at this temperature, the 1200 psi hydraulic pressure shall be rapidly applied 5 times. After each application of pressure, there shall be no evidence of leaks.

4.4.4.5 Dielectric strength. The transmitter shall be subjected to a test voltage of 1,000V (rms) plus twice the working voltage at commercial frequency for a period of 1 minute. The test voltage shall be apllied between the terminals (shorted together) and ground at sea-level atmospheric pressure.

WARNING

The previous involves dangerous voltage which may cause severe shock or injury. Avoid contact with energized components. Remove rings, watches and other metallic objected which may cause shock or burn hazards.

4.4.5 Sampling tests. Four transmitters shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to the following tests.

4.4.5.1 Lot. A lot shall consist of transmitters manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.4.5.2 Failures. When tests are specified on a quantity of transmitters that are selected as representative of a certain lot, and one or more of this number fails to meet the specified tests 2 additional transmitters of the lot represented shall be tested immediately to determine the extent of failure. If one or more transmitters of the additional samples fail to meet the specified tests, the entire lot shall be rejected. Individual performance tests need not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the transmitters.

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4.4.5.3 Special tests. Three transmitters representing the manufacturer's contract or order shall be subjected, to the special tests specified herein. The special tests shall be conducted as follows and in the order specified.

4.4.5.3.1 Test-Group I One of the transmitters shall be subjected to the following tests:

Substitute fuel
Extreme low temperature exposure
Extreme high temperature exposure
Humidity

4.4.5.3.2 Test-Group II. The second transmitter shall be subjected to the following test:

Fungus

4.4.5.3.3 Test-Group III. The third transmitter shall be subjected to the following tests:

Radio interference
Salt fog
Burst pressure

4.4.5.4 Visual examination of product. The transmitter shall be examined to determine its conformance with Section 3 herein, unless such requirements are covered by a test under Section 4.

4.4.5.5 Position error. The Scale error test shall be repeated while the transmitter is in a position rotated 45 and 90 degrees from the normal. The scale errors shall not exceed 2.0 percent of the indicated gallons consumed.

4.4.5.6 Vibration. The transmitter shall be subjected to vibration for a total period of 15 hours on a suitably designed vibration stand. It shall be vibrated for 5 hours in each 3 directions as specified in paragraph covering "Vibration stand". The first hour in each direction shall be at 15 Hz, the second hour 25 Hz, the third 30, the fourth 40, and fifth 50 Hz. The transmitter shall be filled with fuel and capped during this test. Following the vibration period, the transmitter shall be tested for scale errors as specified for the transmitter in the Individual tests. The scale errors shall not exceed the tolerances specified in Table I for the Room temperature scale error tests.

4.4.6 Special tests:

4.4.6.1 Substitute fuel. The transmitter shall be subjected to the Scale error test, except for the 1500 gph test point. The fuel used shall be in accordance with Specification MIL-F-5572. The scale error shall not exceed the tolerances specified in Table I for the Substitute fuel test.

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4.4.6.2 Extreme low temperature exposure. The transmitter shall be placed in a chamber and maintained at a temperature of -65 ±2°C for a period of 48 hours. The cold exposure shall then be continued at -54 ±2°C for an additional period of 24 hours. At the end of the 24-hour period and while the transmitter is still at a temperature of -54 ±2°C the Scale error test as specified in the Low temperature test shall be conducted. The scale errors observed shall not exceed the tolerances specified in Table I. Following the Extreme low temperature exposure test, and after all parts of the transmitter have attained room temperature for at least 1 hour, the transmitter shall be subjected to the Scale error test. The scale error shall not exceed the tolerances specified in Table I. The transmitter shall be inspected for deterioration or damage due to extreme temperature exposure.

4.4.6.3 Extreme high temperature exposure. The transmitter shall be placed in a chamber and maintained at +110 ±2°C for a period of 24 hours. Following this period and while the transmitter is still at +110 ±2°C a Leak test shall be conducted by rapidly applying an hydraulic pressure of 1200 psi 5 times. After each application of pressure, there shall be no evidence of leaks over a period of 5 minutes. Following this period and while the transmitter and the fuel are at a temperature of +55 ±2°C the transmitter shall be subjected to the Scale error test as specified in the High temperature test. The scale errors shall not exceed the tolerances specified in Table I. After the transmitter has attained room temperature for at least 1 hour, the Room temperature scale error test shall be conducted. The scale error shall not exceed the tolerance specified in Table I. The transmitter shall be inspected for deterioration or damage due to extreme temperature exposure.

4.4.6.4 Humidity. The transmitter shall be mounted in the normal operating position and subjected to the following Humidity cycling test. At the completion of the test, the transmitter shall not show signs of corrosion of metal parts or any other damage.

4.4.6.4.1 Humidity cycling. The transmitter shall be placed in a chamber and maintained at a temperature of +65 ±2°C and a relative humidity of 95 percent for a period of 6 hours. The heat shall then be shut off and the transmitter shall then be cooled for 18 hours in this atmosphere in which the humidity shall be allowed to reach 100 percent as the temperature is reduced. (Precautions shall be taken to prevent dripping on the transmitter.) The transmitter shall be subjected to 5 of these humidity cycles. The Scale error test at room temperature shall be conducted at the completion of the cycles. The same tolerances shall apply. All transmitters, subjected to this test, shall be reworked regardless of damage to make them equal to new transmitters.

4.4.7.5 Fungus resistance. The transmitter shall be subjected to fungus resistance tests MIL-STD-810C method 508.2. After this test the transmitter shall be subjected to the Room temperature scale error test. The same tolerances shall apply. There shall be no deterioration as a result of this test which adversely affect subsequent operation.

4.4.7.6 Electromagnetic interference. The electromagnetic tests shall be conducted in accordance with MIL-STD-462 except the frequency range shall be from 0.15 to 150 MHz. The test methods shall be those required by MIL-STD-461 for class A1 category b equipment.

4.4.7.7 Salt fog. The transmitter shall be subjected to salt fog tests MIL-STD-810C method 509.1 for a period of 40 hours. No damage to the transmitter shall result from this test.

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4.4.7.8 Burst pressure. A pressure of 1500 psi shall be applied to the inlet connection with the outlet connection plugged or capped. The transmitter shall withstand this pressure without failure.

4.5 Destroyed parts. All parts, specimens, or assemblies destroyed in making test required by this specification or drawings, to determine compliance with the specification or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

4.6 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each transmitter shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each transmitter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting acitivity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each transmitter shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of transmitters shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Each transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size minimum tare and cube consistent with the protection required.

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5.2.2 Level B. Each transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Level of packaging and packing desired.

d. That invitations to bids, contract or purchase order, will specify that a minimum, of two transmitters will be required as first article samples to be subjected to the first article tests by the contractor to determine compliance with requirements of this specification, and that two transmitters will be required as first article samples to be submitted to the contracting activity.

6.2 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first article production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.3 Subject term (key word) listing.

Fuel flow
Magnetically switched 28 VDC
Totalizing
Transmitter
Type H-1A

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6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

AIR FORCE - 99

Preparing activity:

AIR FORCE - 71

Project number:
6620-F470

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-6368A	Transmitters, True Flow, Totalizing, Type H-1A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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